EC200U&EG91xU Series QuecOpen(SDK) NTP API Reference Manual

LTE Standard Module Series

Version: 1.1

Date: 2023-09-20

Status: Released

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.
Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China
Tel: +86 21 5108 6236
Email: info@quectel.com

Or our local offices. For more information, please visit:
http://www.quectel.com/support/sales.htm.

For technical support, or to report documentation errors, please visit:
http://www.quectel.com/support/technical.htm.
Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an "as available" basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2023. All rights reserved.

About the Document

Revision History

Version	Date	Author	Description
-	2021-03-04	Herry GENG	Creation of the document
1.0	2022-03-28	Charis JIANG	First official release
1.1	2023-09-20	Kruskal ZHU	1. Updated the document name based on the unified QuecOpen naming. 2. Added applicable modules EG912U-GL and EG915U series. 3. Updated the NTP result code enumeration ql_ntp_error_code_e (Chapter 3.3.1.4).

Contents

Table Index

1 Introduction

Quectel EC200U series and EG91xU family (EG912U-GL and EG915U series) modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see ***document [1]***.

This document is applicable to QuecOpen® solution based on CSDK build environment, and introduces the NTP API, related calling process and the example on Quectel EC200U series and EG91xU family modules.

2 NTP API Calling Process

Before enabling NTP to synchronize time, you need to register the network and start a data call to establish a data channel first. If the data channel has been established in other tasks, you can skip the steps of "Wait for network registration" and "Start a data call". The calling process is shown in the figure below:

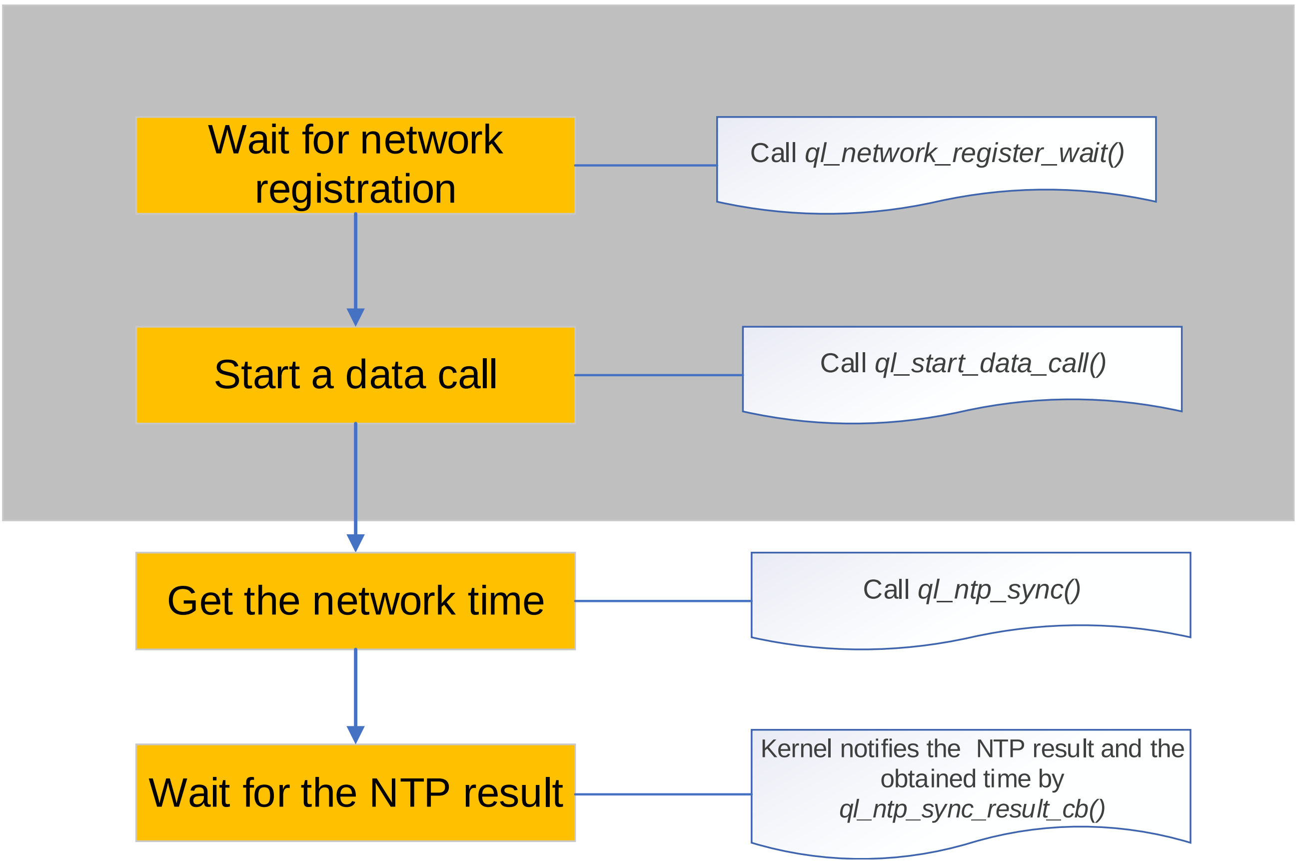

Figure 1: NTP API Calling Process

NOTE

See ***document [2]*** for details about *ql_network_register_wait()* and *ql_start_data_call()*.

3 NTP API

3.1. Header File

ql_ntp_client.h, the header file of NTP API, is in the *\components\ql-kernel\inc* directory. Unless otherwise specified, all header files mentioned in this document are in this directory.

3.2. Example

The NTP example in application layer is provided in the SDK of the module and NTP client example file is in the *components\ql-application\ntp\ntp_demo.c* directory.

3.3. API Description

3.3.1. ql_ntp_sync

This function enables the feature of NTP synchronization time.

- **Prototype**

```
ntp_client_id ql_ntp_sync(const char* host, ql_ntp_sync_option *user_option, ql_ntp_sync_result_cb cb, void *arg, int *error_code)
```

- **Parameter**

host:
[In] The address of NTP server.

user_option:
[In] NTP configuration information. See ***Chapter 3.3.1.2*** for details.

cb:
[In] Callback function for notifying the NTP synchronization result and the obtained time. See ***Chapter 3.3.1.3*** for details.

arg:
[In] The callback argument of the NTP synchronization result notification function.

error_code:
[Out] NTP result code called by the function. See ***Chapter 3.3.1.4*** for details.

- **Return Value**

NTP client handle. See ***Chapter 3.3.1.1*** for details.

3.3.1.1. ntp_client_id

NTP client handle is defined below:

```
typedef int    ntp_client_id
```

3.3.1.2. ql_ntp_sync_option

The structure of NTP configuration information:

```
typedef struct{
        int       pdp_cid;
        int       sim_id;
        int       retry_cnt;
        int       retry_interval_tm;
}ql_ntp_sync_option
```

- **Parameter**

Type	Parameter	Description
int	*pdp_cid*	The PDP channel ID bound to NTP client.
int	*sim_id*	(U)SIM card bound to NTP client.
int	*retry_cnt*	The number of retries when NTP fails to get the time.
int	*retry_interval_tm*	The interval time of NTP request.

3.3.1.3. ql_ntp_sync_result_cb

This function is the event notification function of NTP, notifying the NTP synchronization result and the obtained time.

- **Prototype**

```
typedef void(*ql_ntp_sync_result_cb)(ntp_client_id cli_id, int result, struct tm *sync_time, void *arg)
```

- **Parameter**

cli_id:
[In] NTP client handle, obtained by *ql_ntp_sync()*.

result:
[Out] NTP result code. See ***Chapter 3.3.1.4*** for details.

sync_time:
[Out] Obtained time.

arg:
[In] Callback argument, passed into by *ql_ntp_sync()*.

- **Return Value**

None

3.3.1.4. ql_ntp_error_code_e

The enumeration of NTP result codes:

```
typedef enum
{
QL_NTP_SUCCESS = 0,
QL_NTP_ERROR_UNKNOWN = 550 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_WODBLOCK = 551 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_INVALID_PARAM = 552 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_OUT_OF_MEM = 553 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_SOCKET_ALLOC_FAIL = 554 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_SOCKET_SEND_FAIL = 558 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_SOCKET_RECV_FAIL = 559 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_OPEN_PDP_FAIL = 561 | (QL_COMPONENT_LWIP_QNTP << 16),
QL_NTP_ERROR_DNS_FAIL = 565 | (QL_COMPONENT_LWIP_QNTP << 16),
```

```
QL_NTP_ERROR_TIMEOUT = 569 | (QL_COMPONENT_LWIP_QNTP << 16),
}ql_ntp_error_code_e
```

- **Member**

Member	Description
QL_NTP_SUCCESS	Successful execution.
QL_NTP_ERROR_UNKNOWN	Unknown error(s).
QL_NTP_ERROR_WODBLOCK	NTP request is blocked.
QL_NTP_ERROR_INVALID_PARAM	Invalid parameter(s).
QL_NTP_ERROR_OUT_OF_MEM	Out of memory.
QL_NTP_ERROR_SOCKET_ALLOC_FAIL	Failed to create the socket
QL_NTP_ERROR_SOCKET_SEND_FAIL	Failed to send an NTP request.
QL_NTP_ERROR_SOCKET_RECV_FAIL	Failed to receive an NTP response.
QL_NTP_ERROR_OPEN_PDP_FAIL	Failed to activate PDP.
QL_NTP_ERROR_DNS_FAIL	Failed to parse DNS.
QL_NTP_ERROR_TIMEOUT	NTP request timeout.

4 Appendix References

Table 1: Related Documents

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_CSDK_Quick_Start_Guide
[2] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Data_Call_API_Reference_Manual

Table 2: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
DNS	Domain Name Server
ID	Identifier
IoT	Internet of Things
LTE	Long Term Evolution
NTP	Network Time Protocol
PDP	Packet Data Protocol
SDK	Software Development Kit